

Project Title: Blood Donation Platform - Flutter Application

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1. Abstract

This project involves the development of a mobile application built using Flutter that serves as a platform for blood donation. The main goal is to bridge the gap between blood donors and recipients by providing a convenient and efficient medium for communication. With a clean and user-friendly interface, this application helps users quickly find suitable blood donors or recipients nearby, thus saving crucial time during emergencies.

2. Aim

The aim of this project is to design and develop a cross-platform mobile application that facilitates the process of blood donation by enabling seamless connectivity between donors and recipients. The application aims to reduce the time and effort required to find blood in critical situations.

3. Objectives

- To develop a mobile application using Flutter for both Android and iOS platforms.
 - To allow users to register as either donors or recipients.
 - To implement features to search for blood donors based on blood group and location.
 - To allow real-time communication between users.
 - To ensure user data is securely stored and managed.
 - To provide a responsive and intuitive UI/UX design.
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4. Scope of the Project

The Blood Donation Platform is designed to serve hospitals, emergency services, NGOs, and individuals in need of blood. The app's capabilities can be expanded to include:

- Integration with blood banks and hospitals.
- Scheduling of blood donation camps.
- Notifications and reminders for repeat donations.
- Rewards or recognition for frequent donors.

The application is scalable and adaptable, allowing further development and the addition of advanced features like live donor tracking or analytics for healthcare organizations.

5. Features

- Simple registration and login process.
 - Donor and recipient profile creation and management.
 - Search and filter donors based on blood group and proximity.
 - Contact system via call or email.
 - History of donations and requests.
 - Secure authentication and data management.
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6. Technologies Used

- **Flutter & Dart** for front-end mobile development.
 - **Firebase** for backend services: (not used yet)
 - Firestore (Database)
 - Firebase Authentication
 - **Google Maps API** (Optional) for showing donor locations. (not used yet)
 - **Git & GitHub** for version control.
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7. System Architecture

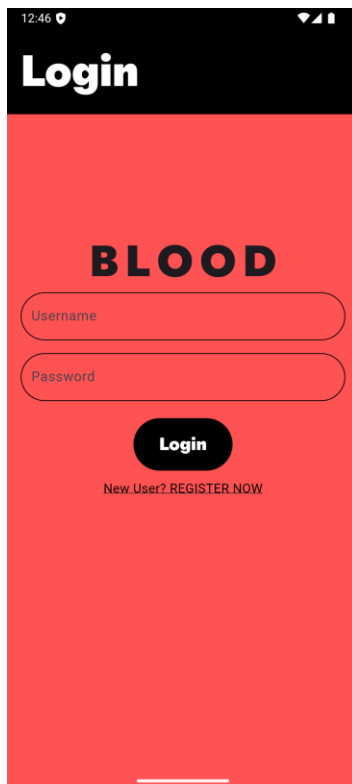
The application follows a client-server architecture:

- The **Flutter** frontend handles the user interface and interactions.
 - **Firebase Authentication** handles secure login/signup.
 - **Cloud Firestore** stores and retrieves user profiles, donation requests, and history.
 - Optional integration with Google Maps provides geolocation-based services.
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8. UI Description

Below are some screenshots of the Blood Donation Platform application demonstrating key features and interface design:

- **Login Screen:** Shows the user login interface with input fields for email and password.
- **Home Screen:** Displays the user dashboard after logging in, with navigation options.
- **Donor List Screen:** A searchable and filterable list of available blood donors.
- **Request Form:** A form for recipients to request blood, including fields for blood group, urgency, and location.
- **User Profile:** Details of donor/recipient with options to update profile information.



9. Conclusion

This project demonstrates how mobile technology can be leveraged to solve a real-world problem in the healthcare sector. By connecting donors and recipients through a digital medium, the application offers a vital service that can save lives. With its scalability and potential for further enhancement, the Blood Donation Platform stands as a promising solution to improve blood donation systems.